

Smart Inventory ERD Initial Design (Updated)

Project Overview

This document contains the updated ERD design for the Smart Inventory & Sales Analytics System project. It also includes the mapping between each table in the project and its corresponding source table from the Wide World Importers (WWI) dataset.

1. Geographic Tables

Countries

Attribute	Type	Description
CountryID (PK)	INT	Unique country identifier
CountryName	NVARCHAR(100)	Country name
Continent	NVARCHAR(100)	Continent name
Region	NVARCHAR(100)	Geographic region

WWI Source

Application.Countries

StateProvinces

Attribute	Type	Description
StateProvinceID (PK)	INT	Unique province identifier
StateProvinceName	NVARCHAR(100)	Province name
CountryID (FK)	INT	Country reference

WWI Source

Application.StateProvinces

Cities

Attribute	Type	Description
CityID (PK)	INT	Unique city identifier

CityName	NVARCHAR(100)	City name
StateProvinceID (FK)	INT	Province reference
Population	BIGINT	City population

WWI Source

Application.Cities

2. Employees

Employees

Attribute	Type	Description
EmployeeID (PK)	INT	Unique employee identifier
FullName	NVARCHAR(100)	Employee full name
IsSalesperson	BIT	Indicates salesperson

WWI Source

Application.People

3. Customer Tables

CustomerCategories

Attribute	Type	Description
CustomerCategoryID (PK)	INT	Customer category identifier
CustomerCategoryName	NVARCHAR(100)	Category name

WWI Source

Sales.CustomerCategories

BuyingGroups

Attribute	Type	Description
BuyingGroupID (PK)	INT	Buying group identifier
BuyingGroupName	NVARCHAR(100)	Buying group name

WWI Source

Sales.BuyingGroups

Customers

Attribute	Type	Description
CustomerID (PK)	INT	Unique customer identifier
CustomerName	NVARCHAR(100)	Customer name
CustomerCategoryID (FK)	INT	Customer category reference
BuyingGroupID (FK)	INT	Buying group reference
CityID (FK)	INT	City reference
DeliveryMethodID	INT	
PhoneNumber	NVARCHAR(30)	Customer phone
WebsiteURL	NVARCHAR(200)	Customer website
CreditLimit	DECIMAL(18,2)	Credit limit
AccountOpenedDate	DATE	Account creation date

WWI Source

Sales.Customers

4. Supplier Tables

SupplierCategories

Attribute	Type	Description
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SupplierCategoryID (PK)	INT	Supplier category identifier
SupplierCategoryName	NVARCHAR(100)	Supplier category name

WWI Source

Purchasing.SupplierCategories

Suppliers

Attribute	Type	Description
SupplierID (PK)	INT	Supplier identifier
SupplierName	NVARCHAR(100)	Supplier name
SupplierCategoryID (FK)	INT	Supplier category reference
CityID (FK)	INT	City reference
DeliveryMethodID	INT	
PhoneNumber	NVARCHAR(30)	Supplier phone
WebsiteURL	NVARCHAR(200)	Supplier website

WWI Source

Purchasing.Suppliers

5. Product Tables

Categories

Attribute	Type	Description
CategoryID (PK)	INT	Product category identifier
CategoryName	NVARCHAR(100)	Product category name

WWI Source

Warehouse.StockGroups

Colors

Attribute	Type	Description
ColorID (PK)	INT	Color identifier
ColorName	NVARCHAR(100)	Product color

WWI Source

Warehouse.Colors

Products

Attribute	Type	Description
ProductID (PK)	INT	Product identifier
ProductName	NVARCHAR(150)	Product name
SupplierID (FK)	INT	Supplier reference
CategoryID (FK)	INT	Product category reference
ColorID (FK)	INT	Product color reference
Brand	NVARCHAR(100)	Product brand
UnitPrice	DECIMAL(18,2)	Product selling price
TaxRate	DECIMAL(5,2)	Product tax rate
QuantityPerOuter	INT	Quantity per package
RecommendedRetailPrice	DECIMAL(18,2)	Suggested retail price

WWI Source

Warehouse.StockItems Warehouse.StockItemStockGroups

6. Inventory Tables

Inventory

Attribute	Type	Description
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ProductID (PK/FK)	INT	Product reference
QuantityOnHand	INT	Current stock quantity
ReorderLevel	INT	Minimum stock level
TargetStockLevel	INT	Target stock level
LastCostPrice	DECIMAL(18,2)	Last purchase cost

WWI Source

Warehouse.StockItemHoldings

InventoryTransactions

Attribute	Type	Description
TransactionID (PK)	INT	Inventory transaction identifier
ProductID (FK)	INT	Product reference
TransactionTypeID (FK)	INT	Transaction type reference
CustomerID (FK)	INT	Customer reference
SupplierID (FK)	INT	Supplier reference
Quantity	INT	Transaction quantity
TransactionDate	DATETIME	Transaction date

WWI Source

Warehouse.StockItemTransactions

7. Lookup Tables

PaymentMethods

Attribute	Type	Description
PaymentMethodID (PK)	INT	Payment method identifier
PaymentMethodName	NVARCHAR(100)	Payment method name

WWI Source

Application.PaymentMethods

DeliveryMethods

Attribute	Type	Description
DeliveryMethodID (PK)	INT	Delivery method identifier
DeliveryMethodName	NVARCHAR(100)	Delivery method name

WWI Source

Application.DeliveryMethods

TransactionTypes

Attribute	Type	Description
TransactionTypeID (PK)	INT	Transaction type identifier
TransactionTypeName	NVARCHAR(100)	Transaction type name

WWI Source

Application.TransactionTypes

8. Sales Tables

Orders

Attribute	Type	Description
OrderID (PK)	INT	Order identifier
CustomerID (FK)	INT	Customer reference
EmployeeID (FK)	INT	Salesperson reference
PaymentMethodID (FK)	INT	Payment method reference
DeliveryMethodID (FK)	INT	Delivery method reference
OrderDate	DATE	Order date

ExpectedDeliveryDate	DATE	Expected delivery date
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WWI Source

Sales.Orders

OrderDetails

Attribute	Type	Description
OrderDetailID (PK)	INT	Order detail identifier
OrderID (FK)	INT	Order reference
ProductID (FK)	INT	Product reference
Quantity	INT	Ordered quantity
UnitPrice	DECIMAL(18,2)	Unit selling price
TaxRate	DECIMAL(5,2)	Tax rate
ExtendedPrice	DECIMAL(18,2)	Total line price
LineProfit	DECIMAL(18,2)	Profit per line

WWI Source

Sales.OrderLines Warehouse.StockItemHoldings

Revenue

ExtendedPrice =Quantity×UnitPrice

Cost

Cost=Quantity×LastCostPrice(Warehouse.StockItemHoldings)

Profit

LineProfit =Revenue–Cost

9. Financial Transactions

CustomerTransactions

Attribute	Type	Description
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CustomerTransactionID (PK)	INT	Customer transaction identifier
CustomerID (FK)	INT	Customer reference
TransactionTypeID (FK)	INT	Transaction type reference
PaymentMethodID (FK)	INT	Payment method reference
TransactionDate	DATE	Transaction date
AmountExcludingTax	DECIMAL(18,2)	Amount before tax
TaxAmount	DECIMAL(18,2)	Tax amount
TransactionAmount	DECIMAL(18,2)	Final transaction amount
OutstandingBalance	DECIMAL(18,2)	Remaining balance

WWI Source

Sales.CustomerTransactions

SupplierTransactions

Attribute	Type	Description
SupplierTransactionID (PK)	INT	Supplier transaction identifier
SupplierID (FK)	INT	Supplier reference
TransactionTypeID (FK)	INT	Transaction type reference
PaymentMethodID (FK)	INT	Payment method reference
TransactionDate	DATE	Transaction date
TransactionAmount	DECIMAL(18,2)	Final transaction amount
OutstandingBalance	DECIMAL(18,2)	Remaining balance

WWI Source

Purchasing.SupplierTransactions

Geographic Relationships

Parent Table	Relationship	Child Table
Countries	Has	StateProvinces
StateProvinces	Has	Cities
Cities	Contains	Customers
Cities	Contains	Suppliers

Core Business Relationships

Parent Table	Relationship	Child Table
CustomerCategories	Has	Customers
BuyingGroups	Contains	Customers
SupplierCategories	Has	Suppliers
Categories	Contains	Products
Colors	Defines	Products
Suppliers	Supplies	Products
Products	Has	Inventory
Customers	Place	Orders
Employees	Handle	Orders
Orders	Contain	OrderDetails
Products	Referenced In	OrderDetails
Products	Have	InventoryTransactions
TransactionTypes	Define	InventoryTransactions
Customers	Have	CustomerTransactions

Suppliers

Have

SupplierTransactions

Final Notes

This model represents a complete Retail & Supply Chain Analytics System. The design supports:

- Inventory Analysis
- Sales Analytics
- Customer Analysis
- Supplier Performance
- Financial Transactions
- Geographic Analysis
- Supply Chain Monitoring
- Power BI Dashboards
- Data Warehouse Implementation
- ETL Process using SSIS